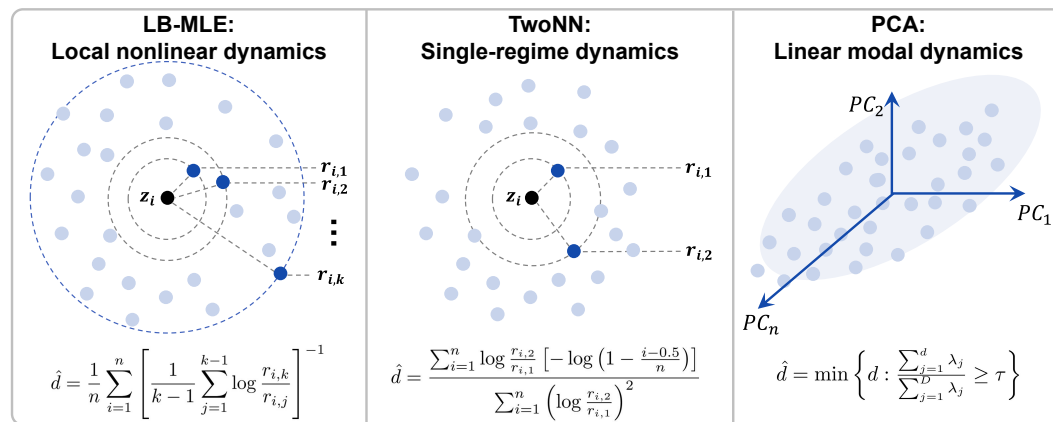
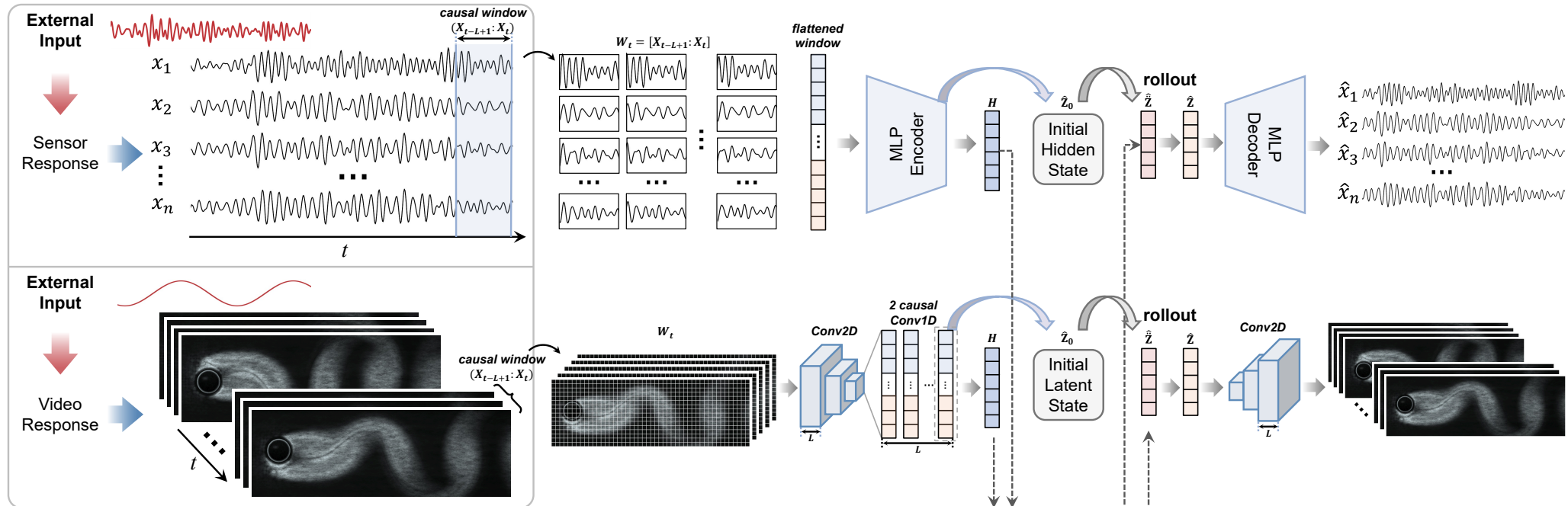
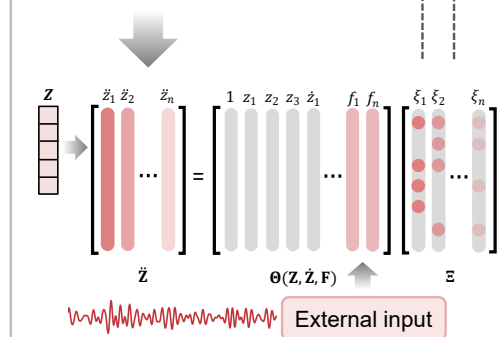
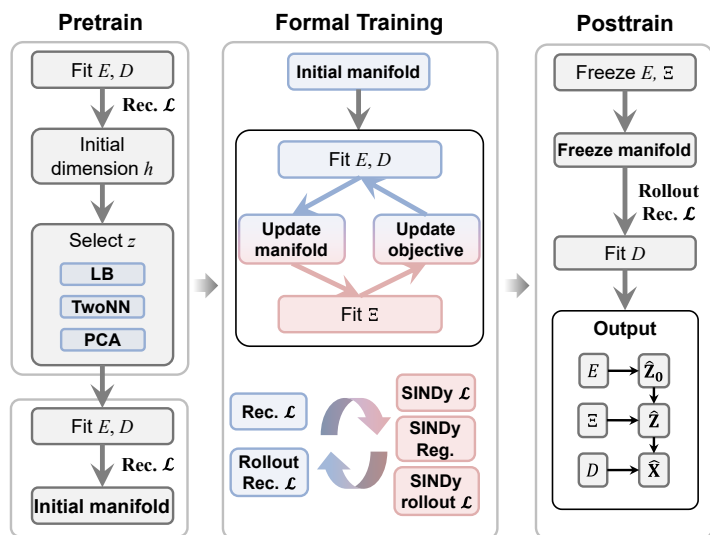


a

Estimation of z Dimensions



$$\begin{aligned} \mathcal{L} = & \lambda_1 \|x - D(z^E)\|_2^2 && \text{Rec. } \mathcal{L} \\ & + \lambda_2 \|x - D(z^R)\|_2^2 && \text{Rollout Rec. } \mathcal{L} \\ & + \lambda_3 \|z^R - \text{sg}(z^E)\|_2^2 && \text{SINDy rollout } \mathcal{L} \\ & + \lambda_4 \|\ddot{z}^E - \Theta \Xi_m\|_2^2 && \text{SINDy } \mathcal{L} \\ & + \lambda_5 \|\Xi_m\|_1 && \text{SINDy Reg.} \end{aligned}$$

b**c**